

Andrew Miller

BACKEND SOFTWARE ENGINEER

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SUMMARY

Backend software engineer with approximately 4.5 years of professional C#/.NET experience, focused on modernizing legacy systems. For 3.5 years, served as the primary backend developer for a compliance-intensive mortgage document generation platform where precision is critical. Outside of work, designs, builds, and runs atmiller.org end to end, a Rust web application on infrastructure defined in C# with Pulumi (see Projects).

SKILLS

Languages	C#, Rust, SQL/T-SQL, JavaScript, HTML/CSS
.NET & Web	ASP.NET Core, .NET 8-10, Blazor (Server & WebAssembly), Entity Framework Core, Dapper, MediatR/CQRS, REST/GraphQL APIs, OpenAPI/Swagger, Postman, .NET Aspire
Rust & Web	Leptos (SSR + WebAssembly hydration), Axum, Tokio, Tower, rustls, tracing, cargo-deny
Data & Messaging	SQL Server/Azure SQL, Redis, RabbitMQ (MassTransit), event-driven architecture
Cloud & DevOps	Azure, DigitalOcean, Pulumi, GitHub Actions, CI/CD, containers (Docker, Podman), OpenTelemetry, Git
Testing	Unit (xUnit/bUnit) & end-to-end (Playwright) testing
AI	LLM feature integration (Gemini), agentic development workflows (Claude Code)
Mortgage Domain	Encompass (ICE Mortgage) LOS integration, MISMO/UCD 2 XML, TRID (Loan Estimate & Closing Disclosure), RESPA, compliant document generation (Aspose)

PROJECTS

Personal Site - Rust web application, built and operated end to end Jul 2026 - present

atmiller.org · github.com/atm1150/personal-site

- Built a server-rendered Rust web application with Leptos and Axum, hydrated with WebAssembly.
- Hardened the server with a Content Security Policy and full security-header suite, per-client rate limiting, and request, body, and header-read limits.
- Terminated TLS in the server itself with rustls, with certificates issued and renewed automatically through DNS-01.
- Gated every pull request on formatting, lint, dependency audit, and tests; on merge, CI builds, scans, and publishes the container image and deploys it over SSH, with a tagged rollback workflow.
- Provisioned the host, reserved IP, and firewall as code with Pulumi in C#, with unit tests over the stack configuration.
- Instrumented the server with OpenTelemetry traces and logs, exported to a .NET Aspire orchestrator that runs the full stack locally.

EXPERIENCE

PPDocs - Software Developer, C#/.NET Backend Jan 2023 - Jun 2026

Mortgage loan document-generation platform serving lenders in all 50 states

- Modernized a 20-year-old VB.NET codebase to .NET 8-10, migrating business logic from legacy stored procedures into MediatR/CQRS handler pipelines using Entity Framework Core and Dapper.
- Built a multithreaded C# tool that parsed a 50+ page legacy XHTML/JavaScript order form governed by state-specific legal, internal, and monetary rules, then generated equivalent Blazor Server pages with visibility conditions, save/retrieval wiring, and backing C# behavior objects, automating the bulk of the UI migration.

- Designed and implemented the team's .NET Aspire setup end to end, orchestrating five services plus Azure SQL, RabbitMQ, Redis, and MailHog under one local startup, with a test-database builder for reproducible resets.
- Wrote a custom Azure SQL Aspire integration to keep local development from falling back to the default SQL Server (MSSQL) container, which cannot run on the team's ARM-based laptops.
- Built event-driven workflows that route newly submitted loans from the order form UI or API to the intranet whiteboard for attorney review.
- Built mortgage document-generation features, assembling C#-computed content and tables into Aspose Word templates (rendered to PDF) - a library of 100+ compliant loan documents spanning TRID, RESPA, and non-RESPA packages.
- Built an LLM-powered mapping feature using Gemini Flash to match customers' free-text interest-rate source descriptions to platform index definitions and present the suggested match for confirmation, automating a manual, error-prone step.
- Built the Encompass Partner Connect integration for ICE Mortgage Technology and implemented UCD 2 delivery in MISMO XML to clients or directly to the GSEs, including Fannie Mae and Freddie Mac.

Aeravida - Software Developer, .NET
Jan 2022 - Dec 2022

E-commerce platform for handmade and fair-trade goods

- Helped rebuild a legacy .NET e-commerce application on ASP.NET MVC while maintaining and extending the live production site.
- Integrated Wayfair's GraphQL and Walmart's REST seller APIs to publish and sync the product catalog, opening two additional marketplace sales channels.
- Recruited after a year by the company's co-owner - PPDocs' system architect - to join PPDocs as a backend developer.

Silgan Closures - Operator to Mechanic (prior to software career)
May 2014 - Jan 2022

- Progressed from machine operator to mechanic, diagnosing and repairing faults on production machinery.
- Performed troubleshooting and quality-assurance checks at both levels - the same systematic diagnose-isolate-fix loop later applied to software debugging.

EDUCATION & TRAINING

The Software Guild - C#/.NET Full-Stack Developer Certification	2020 - 2021
University of Evansville - Coursework in Mathematics, Physics & Accounting	2010 - 2014